Constructive method on the strong Goldbach conjecture yielding various conditional and unconditional results

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Abstract

In the present paper we formalize a constructive method that generates an infinite family of Goldbach pairs (p, q) with $p + q = 2n$. We first establish a fundamental discrete counting bound giving a guaranteed Goldbach density of $O(n^{-\theta^2})$. We then prove full conditional completeness under the Cramér–Granville conjecture above a phase transition at 10^{69}. By analyzing the bounding constraints, we derive a critical gap hypothesis of $O(x^{\sqrt{\log_x 2}})$. Using established results on the distribution of prime numbers, we prove unconditionally that the density of non-representable even numbers approaches 0 with explicit rate $e^{-0.5\sqrt{\log x}}$ via chain breaks. Finally, we establish an optimal conditional complete-coverage theorem under our newly derived gap hypothesis.

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1 Construction Method and Prime Distribution Bounds

Let $(n, p, q, p', q', c) \in (\mathbb{N}^*)^6$ with p, q prime such that $p + q = 2n$, and p', q' prime such that $p' + q' = 2n + 2c$. Let $k \in \mathbb{Z}$. We set

$$p' = p + 2k, \qquad q' = q - 2k + 2c.$$

Since p and p' are both odd primes (for n large enough), $p' = p + 2k$ is automatically odd for any integer k, so parity is consistent throughout. Similarly $q' = q - 2k + 2c$ is odd for any integer k and positive integer c.

We utilize the following established prime distribution result:

Hoheisel–Baker–Harman–Pintz. There exists $\theta \in (1/2, 1)$ such that for all sufficiently large $m \in \mathbb{N}$ there always exists a prime in $(m,\ m+2m^\theta]$ and a prime in $[m-2m^\theta,\ m)$. We use intervals of length $2m^\theta$ (rather than m^θ) to guarantee the existence of a prime of prescribed parity within the interval. The best unconditional value is $\theta = 0.525$ (Baker–Harman–Pintz, 2001); conditionally under RH one may take $\theta = 0.5 + \varepsilon$.

Applying this to our construction, there exists a prime p' with

$$p' \in \big(2n - 5 - 2(2n-5)^\theta,\ 2n - 5\big],$$

which gives

$$\frac{q - 5 - 2(2n-5)^\theta}{2} < k \leq \frac{q-5}{2}.$$

Lemma 1.1. *$p' = p + 2k$ can be prime for $k \in \left(\frac{q - 5 - 2(2n-5)^\theta}{2},\ \frac{q-5}{2}\right]$. The existence of such a prime p' is guaranteed by the Hoheisel–BHP result.*

There also exists a prime q' with $q' \in (q-2k,\ q-2k+2(q-2k)^\theta]$, giving $0 < c \leq (q-2k)^\theta$.

Lemma 1.2. *$q' = q - 2k + 2c$ can be prime for $c \in (0,\ (q-2k)^\theta]$, as long as $(q-2k)^\theta \geq 1$, which holds for all large enough n.*

Studying the worst-case for c:

$$c_{\max} = (q - 2k_{\min})^\theta = \big(5 + 2(2n-5)^\theta\big)^\theta \underset{n\to\infty}{\sim} 2^\theta \cdot (2n)^{\theta^2}.$$

Lemma 1.3. *There exists a pair of primes (p', q') constructed from (p, q) with $p + q = 2n$ such that $p' + q' = 2n + 2c$ with $c \in (0,\ 2^\theta \cdot (2n)^{\theta^2}]$.*

2 Fundamental Counting and Density Bound

Let a count the successive Goldbach evens $2n_i$ guaranteed by our method up to $2n$. For a given interval, the maximum discrete step between consecutive guaranteed evens is bounded by

$$\Delta_{\max} = 2c_{\max} \approx 2^{\theta+1} \cdot (2n)^{\theta^2}.$$

Summing these maximum discrete steps up to $2n$, the minimum number of guaranteed valid Goldbach evens a satisfies:

$$a \cdot \Delta_{\max} \geq 2n \implies a \cdot 2^{\theta+1} \cdot (2n)^{\theta^2} \geq 2n.$$

Therefore, the counting formula for the guaranteed representations is:

$$a \geq C \cdot (2n)^{1-\theta^2}, \qquad \text{where } C = 2^{-(\theta+1)}.$$

The guaranteed density $d(2n)$ of Goldbach representations derived from this discrete construction is:

$$d(2n) = \frac{a}{2n} \geq C \cdot (2n)^{-\theta^2}.$$

Theorem 2.1. *Up to a given $2n$, the number of valid Goldbach evens guaranteed by the construction satisfies $a \geq C \cdot (2n)^{1-\theta^2}$.*

- *Under the unconditional bound $\theta = 0.525$: $a \geq C \cdot (2n)^{0.724375}$.*
- *Under RH ($\theta \to 0.5$): $a \geq C \cdot (2n)^{0.75}$.*

3 Completeness under Cramér–Granville

Under Cramér–Granville, the gap between consecutive primes near x is at most $O(\ln^2 x)$. Translating to our interval form $2x^\theta = O(\ln^2 x)$ gives $\theta = (2 \ln \ln x)/\ln x \to 0$ as $x \to \infty$.

3.1 The Phase Transition at $n \approx 10^{69}$

Lemma 3.1. *Under Cramér–Granville, for all $n > 10^{69}$, the only admissible integer value consistent with Lemma 3 is $c = 1$.*

Proof. Under Cramér–Granville the effective θ satisfies $\theta = 2\ln(\ln(2n))/\ln(2n)$. The upper bound on $2c_{\max}$ from Lemma 3 is $2^{\theta+1} \cdot (2n)^{\theta^2}$. Direct computation shows this quantity drops below 4 at $n \approx 10^{68}$–10^{69}. Specifically, at $n = 10^{69}$:

$$2c_{\max} \approx 3.984 < 4.$$

Since c must be a positive integer with $0 < c \leq (q-2k)^\theta$ and $(q-2k)^\theta < 2$ for all $n > 10^{69}$, the only admissible value is $c = 1$. Lemma 2 guarantees $c = 1$ is always achievable since $(q-2k)^\theta \geq 1$. □

Lemma 3.2. *Under Cramér–Granville, if $2n$ has a Goldbach representation for some $n > 10^{69}$, then $2n+2$ also has a Goldbach representation.*

Proof. By our construction $c = 1$ produces a valid Goldbach pair (p', q') with $p' + q' = 2n+2$. Existence is guaranteed by the BHP result under Cramér–Granville. The step is deterministic and unbroken. □

3.2 Conditional Theorem

Theorem 3.3 (Full Conditional Goldbach)**.** *Assume:*

(H1) Cramér–Granville's conjecture: the gap between consecutive primes near x is at most $O(\ln^2 x)$.

(H2) Every even number in $(4 \times 10^{18},\ 10^{69})$ has a Goldbach representation.

Then every even number greater than 4 has a Goldbach representation.

Proof. Oliveira e Silva, Herzog and Pardi verify every even up to 4×10^{18}. By (H2), every even in $(4 \times 10^{18}, 10^{69})$ is Goldbach. So every even up to 10^{69} is Goldbach, establishing the induction base. For $n > 10^{69}$, Lemma 5 gives the inductive step. The result follows by induction. □

4 Derivation of the Critical Gap Function

We seek the most relaxed gap hypothesis that still guarantees complete coverage. The condition $2c_{\max} < 4$ becomes, after taking logarithms:

$$\theta^2 \ln(2n) + \theta \ln 2 - \ln 2 < 0.$$

The positive root of $\theta^2 L + \theta \ln 2 - \ln 2 = 0$ (where $L = \ln(2n)$) is

$$\theta_{\text{crit}}(n) = \frac{\sqrt{\ln 2\,(4\ln(2n) + \ln 2)} - \ln 2}{2\ln(2n)} \sim \sqrt{\frac{\ln 2}{\ln(2n)}} = \sqrt{\log_{2n} 2}.$$

The corresponding critical gap function is therefore:

$$G_{\text{crit}}(x) = O\left(x^{\sqrt{\log_x 2}}\right).$$

Under this hypothesis, the effective θ satisfies $\theta = \sqrt{\ln 2/\ln(2n)} - \ln 2/\ln(2n)$. Substituting into $\log(2c_{\max}) = (\theta + 1)\ln 2 + \theta^2 \ln(2n)$ gives

$$\log(2c_{\max}) = 2\ln 2 - \frac{(\ln 2)^{3/2}}{\sqrt{\ln(2n)}} < 2\ln 2 = \log 4$$

for all finite n, so $2c_{\max} < 4$ always. Furthermore $2c_{\max} \to 4$ from below, with gap $(\ln 2)^{3/2}/\sqrt{\ln(2n)} > 0$ for all finite n.

5 Unconditional Density via Chain Breaks

5.1 Exceptional Set Setup

Let $G(x) = e^{0.5\sqrt{\log x}}$. Define the exceptional set

$$\mathcal{E}(x, G) = \#\{p \le x \text{ prime} : \text{gap}(p) > G(p)\}.$$

The standard exceptional set bound gives

$$\mathcal{E}(x, G) \le \frac{x}{G(x)} = \frac{x}{e^{0.5\sqrt{\log x}}}.$$

5.2 Chain Breaks

A *chain break* at $2n$ occurs when the construction fails to produce a valid Goldbach pair for $2n$ from any verified Goldbach even below $2n$. Chain breaks split into two types.

P-side failure. No prime p' exists in $(p, p + 2G(p)]$, i.e. the gap after p exceeds $G(p)$ for some prime $p \leq x$.

Q-side failure. For some admissible k, no prime q' exists in $(q-2k,\ q-2k+2G(q-2k)]$, i.e. the gap after $q-2k$ exceeds $G(q-2k)$.

Lemma 5.1 (P-side bound)**.** *The number of p-side failures up to x is at most $\mathcal{E}(x, G) \leq x/G(x)$.*

Proof. A p-side failure at $2n$ requires a gap after a prime $p \leq x$ exceeding $G(p)$. Each such prime contributes at most one failure. The count is therefore at most $\mathcal{E}(x, G) \leq x/G(x)$. ☐

Lemma 5.2 (Q-side bound)**.** *The number of q-side failures up to x is at most $\mathcal{E}(x^\theta, G) \leq x^\theta/e^{0.5\sqrt{\theta \log x}}$.*

Proof. From Lemma 1, the admissible values of k satisfy $k \in \left(\frac{q-5-2(2n-5)^\theta}{2},\ \frac{q-5}{2}\right]$, so

$$q - 2k \in \left[5,\ 5 + 2(2n-5)^\theta\right].$$

For $n \leq x$, we have $q - 2k \leq 5 + 2(2n)^\theta \leq x^\theta$ for large x. A q-side failure therefore requires a gap after some prime $p \leq x^\theta$ exceeding $G(p)$. The count is at most $\mathcal{E}(x^\theta, G) \leq x^\theta/G(x^\theta)$. ☐

Theorem 5.3 (Unconditional Goldbach Density)**.** *The density of even numbers up to x that are* not *guaranteed Goldbach by the construction is at most*

$$\frac{1}{e^{0.5\sqrt{\log x}}} \longrightarrow 0 \quad \textit{unconditionally as } x \to \infty,$$

faster than any power of $\log x$. *In particular* $\lim_{x\to\infty} d(x) = 1$ *unconditionally, where* $d(x)$ *denotes the density of Goldbach-representable even numbers up to x.*

Proof. Every chain break is either a p-side or q-side failure. By the two lemmas:

$$\#\{\text{chain breaks up to } x\} \leq \frac{x}{G(x)} + \frac{x^\theta}{G(x^\theta)} = \frac{x}{e^{0.5\sqrt{\log x}}} + \frac{x^\theta}{e^{0.5\sqrt{\theta \log x}}}.$$

Since $x^\theta/e^{0.5\sqrt{\theta \log x}} = o\left(x/e^{0.5\sqrt{\log x}}\right)$, the p-side dominates and the total is $O(x/G(x))$. The number of even integers up to x is $x/2$, so the exception density is at most

$$\frac{2}{G(x)} = \frac{2}{e^{0.5\sqrt{\log x}}} \to 0.$$

☐

5.3 Comparison with Best Known Results

The unconditional exceptional set $E(x) = \#\{\text{even } n \le x : n \ne p + q\}$ has been studied extensively via circle method and zero-density techniques. Montgomery and Vaughan (1975) first established $E(x) \ll x^{1-\delta}$ for some unspecified $\delta > 0$. Subsequent work made δ explicit: Chen and Pan gave $\delta > 0.01$; Hongze Li (2000) reached $E(x) = O(x^{0.914})$; and the current best is due to Pintz (2018), who achieved $E(x) = O(x^{0.72})$, i.e. $\delta = 0.28$.

Our result gives exception density $O\Big(e^{-0.5\sqrt{\log x}}\Big)$, equivalently $E(x) = O\Big(x/e^{0.5\sqrt{\log x}}\Big)$. Since $e^{0.5\sqrt{\log x}} = o(x^{\varepsilon})$ for any $\varepsilon > 0$, this bound is asymptotically weaker than Pintz for large x. The contribution of Theorem 5.3 is therefore not in the asymptotic rate, but in the method: the bound arises from a direct constructive argument in which chain breaks are the sole identified source of exceptions, and every component of the bound is explicit. Any improvement to the Hoheisel exponent below 0.525 directly and transparently improves the q-side bound by the same amount.

6 The Optimal Conditional Theorem

6.1 Comparison with Cramér–Granville

At $\log_{10} x = 68$: Cramér–Granville gives gap ≈ 24516, our derived critical gap hypothesis gives ≈ 16724 (ours is stronger). At $\log_{10} x = 80$: the two are approximately equal (crossing point). At $\log_{10} x = 100$: Cramér–Granville gives ≈ 53019, ours gives ≈ 153317 (ours is weaker). The two hypotheses are therefore *incomparable*: neither implies the other.

6.2 Optimal Conditional Theorem

Theorem 6.1 (Optimal Conditional Goldbach)**.** *Assume:*

(H1) There exists an explicitly verified x_0 such that for all $x > x_0$,* $\text{gap}(x) \le x^{\sqrt{\log_x 2}}$.

(H2) Every even number in $(4 \times 10^{18},\ 2x_0)$ has a Goldbach representation.*

Then every even number greater than 4 has a Goldbach representation.

Proof. Oliveira e Silva et al. verify every even up to 4×10^{18}. By (H2*), every even in $(4 \times 10^{18}, 2x_0)$ is Goldbach. So every even up to $2x_0$ is Goldbach. For $n > x_0$, H1* forces $\log(2c_{\max}) < 2\ln 2$ strictly, so $2c_{\max} < 4$ and the only admissible integer value is $c = 1$. By Lemmas 1 and 2, the construction produces a valid pair (p', q') with $p' + q' = 2n + 2$. By induction every even greater than 4 is Goldbach. □

Remark on optimality. H1* is asymptotically weaker than Cramér–Granville, making Theorem 6.1 a genuinely more minimal conditional result than Theorem 3.3. We state as an open problem whether H1* can be established unconditionally or derived from existing conjectures in prime gap theory.

6.3 Complementarity of the Main Results

Theorem 5.3 establishes unconditionally that almost all even numbers are Goldbach in the density sense. What it does not establish is that the exception set is empty, only that it has density zero. Closing this gap is precisely what Theorem 6.1 accomplishes under H1*. The two results are complementary: Theorem 5.3 is unconditional and gives density 1; Theorem 6.1 is conditional on a single derived gap hypothesis and gives complete coverage.

7 Open Problems and Future Directions

The constructive framework formalized in this paper introduces a dynamic shifting machinery that opens several clear avenues for optimization and further study. We propose the following open problems to the mathematical community:

7.1 Probabilistic Refinement of Chain-Break Concurrence

The unconditional exception density of $O\left(e^{-0.5\sqrt{\log x}}\right)$ established in Theorem 5.3 is dictated by a strict geometric union bound over the exceptional set $\mathcal{E}(x, G)$. However, a true structural chain break at $2n$ requires a simultaneous failure of candidate primes across both the p-side and q-side intervals. Under standard heuristics regarding the local independence of prime gap distributions, the probability of such mutual exclusion should decay superexponentially. Can a formal probabilistic framework (e.g., extensions of Gallagher's singular series or short-interval random models) be utilized to rigorously tighten the chain-break bound?

7.2 Theoretical Constraints on the Critical Gap Function

The derived critical gap hypothesis, $G_{\text{crit}}(x) = O\left(x^{\sqrt{\log_x 2}}\right)$, serves as the exact operational threshold for complete coverage ($c = 1$) in our construction. While we have demonstrated that this hypothesis is asymptotically weaker than the Cramér–Granville conjecture, its unconditional validity remains open. We raise the question of whether H_1^* can be derived from classical zero-density estimates of zeta-zeros, or if a counterexample can be constructed using Maier-type matrix configurations in short intervals.

7.3 Average-Case Optimization of the Discrete Step Size

The fundamental counting bound $a \geq C \cdot (2n)^{1-\theta^2}$ established in Section 2 utilizes a strict maximum step constraint $\Delta_{\max} = 2c_{\max}$ to ensure deterministic containment. Empirically, the actual required shift $2c$ to locate the next valid Goldbach pair is substantially smaller than this upper limit. Investigating the distribution of c as a localized arithmetic function could yield an average-case step size $\langle 2c \rangle \ll \Delta_{\max}$, thereby significantly lifting the absolute multiplier constant C.

7.4 Application to General Linear Additive Frameworks

The algebraic structure of the transformation equations, $p' = p + 2k$ and $q' = q - 2k + 2c$, relies on preserving the invariant parity sum $2n$. An intriguing direction is whether this exact shifting chain can be generalized to establish bounded exceptional sets for other binary or ternary problems, such as Polignac's conjecture on prime gaps or localized variations of the twin prime problem.———

7.5 And much more can be done to improve on this.

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